

Question ID 128c75e2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 128c75e2

The function g is defined by $g(x) = \frac{|x|}{a} - 14$, where $a < 0$. What is the product of $g(15a)$ and $g(7a)$?

ID: 128c75e2 Answer

Correct Answer:

609

Rationale

The correct answer is 609. It’s given that the function g is defined by $gx = \frac{x}{a} - 14$, where $a < 0$. Substituting $15a$ for x in function g yields $g15a = \frac{15a}{a} - 14$. This function can be rewritten as $g15a = \frac{15a}{a} - 14$, or $g15a = 15\frac{a}{a} - 14$. Since $a < 0$, it follows that $\frac{a}{a} = -1$. Substituting -1 for $\frac{a}{a}$ in $g15a = 15\frac{a}{a} - 14$ yields $g(15a) = 15(-1) - 14$, or $g(15a) = -29$. Similarly, substituting $7a$ for x in function g yields $g7a = \frac{7a}{a} - 14$. This function can be rewritten as $g7a = \frac{7a}{a} - 14$, or $g7a = 7\frac{a}{a} - 14$. Since $a < 0$, it again follows that $\frac{a}{a} = -1$. Substituting -1 for $\frac{a}{a}$ in $g7a = 7\frac{a}{a} - 14$ yields $g(7a) = 7(-1) - 14$, or $g(7a) = -21$. Therefore, $g(15a) = -29$ and $g(7a) = -21$. Thus, the product of $g(15a)$ and $g(7a)$ is $(-29)(-21)$, or 609.

Question Difficulty:

Hard

Question ID ebed7dc6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: ebed7dc6

An auditorium has seats for **1,800** people. Tickets to attend a show at the auditorium currently cost **\$4.00**. For each **\$1.00** increase to the ticket price, **100** fewer tickets will be sold. This situation can be modeled by the equation $y = -100x^2 + 1,400x + 7,200$, where x represents the increase in ticket price, in dollars, and y represents the revenue, in dollars, from ticket sales. If this equation is graphed in the xy -plane, at what value of x is the maximum of the graph?

- A. 4
- B. 7
- C. 14
- D. 18

ID: ebed7dc6 Answer

Correct Answer:

B

Rationale

Choice B is correct. It’s given that the situation can be modeled by the equation $y = -100x^2 + 1,400x + 7,200$, where x represents the increase in ticket price, in dollars, and y represents the revenue, in dollars, from ticket sales. Since the coefficient of the x^2 term is negative, the graph of this equation in the xy -plane opens downward and reaches its maximum value at its vertex. If a quadratic equation in the form $y = ax^2 + bx + c$, where a , b , and c are constants, is graphed in the xy -plane, the x -coordinate of the vertex is equal to $-\frac{b}{2a}$. For the equation $y = -100x^2 + 1,400x + 7,200$, $a = -100$, $b = 1,400$, and $c = 7,200$. It follows that the x -coordinate of the vertex is $-\frac{1,400}{2(-100)}$, or 7. Therefore, if the given equation is graphed in the xy -plane, the maximum of the graph occurs at an x -value of 7.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Question ID 722de804

Assessment	Test	Domain	Skill	Difficulty	ID: 722de804
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	What is the sum of the solutions to the given equation?	$(x - 47)^2 = 1$

ID: 722de804 Answer

Correct Answer:

94

Rationale

The correct answer is 94. Taking the square root of each side of the given equation yields $x - 47 = 1$ or $x - 47 = -1$. Adding 47 to both sides of the equation $x - 47 = 1$ yields $x = 48$. Adding 47 to both sides of the equation $x - 47 = -1$ yields $x = 46$. Therefore, the sum of the solutions to the given equation is $48 + 46$, or 94.

Question Difficulty:

Hard

Question ID 35e05e19

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 35e05e19

A park ranger hung squirrel houses each in the shape of a right rectangular prism for fox squirrels. Each house has a height of **11** inches. The length of each house's base is x inches, which is **1** inch more than the width of the house's base. Which function V gives the volume of each house, in cubic inches, in terms of the length of the house's base?

- A. $V(x) = 11x(x - 1)$
- B. $V(x) = 11x(x + 1)$
- C. $V(x) = x(x + 11)(x - 1)$
- D. $V(x) = x(x + 11)(x + 1)$

ID: 35e05e19 Answer

Correct Answer:

A

Rationale

Choice A is correct. The volume of a prism is equal to the area of its base times its height. It's given that the length of each house's base is x inches and that this length is 1 inch more than the width, in inches, of the house's base. It follows that the width, in inches, of the house's base is $x - 1$. The area of a rectangle is the product of its length and its width. Therefore, the area of the base of the house is $x(x - 1)$ square inches. It's given that the height of each house is 11 inches. Therefore, the function V that gives the volume of each house, in cubic inches, in terms of the length of the house's base x is $V(x) = x(x - 1)11$, or $V(x) = 11x(x - 1)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Question ID 88a0c425

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	■ ■ ■

ID: 88a0c425

$$-2x^2 + 20x + c = 0$$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c ?

- A. **-68**
- B. **-50**
- C. **-32**
- D. **0**

ID: 88a0c425 Answer

Correct Answer:
B

Rationale

Choice B is correct. It's given that the equation $-2x^2 + 20x + c = 0$, where c is a constant, has exactly one solution. A quadratic equation of the form $ax^2 + bx + c = 0$ has exactly one solution if and only if its discriminant, $b^2 - 4ac$, is equal to zero. It follows that for the given equation, $a = -2$ and $b = 20$. Substituting -2 for a and 20 for b in $b^2 - 4ac$ yields $20^2 - 4(-2)(c)$, or $400 + 8c$. Since the discriminant must equal zero, it follows that $400 + 8c = 0$. Subtracting 400 from both sides of this equation yields $8c = -400$. Dividing each side of this equation by 8 yields $c = -50$. Therefore, the value of c is -50 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Hard

Question ID 8462b105

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 8462b105

The function f gives the product of a number, x , and a number that is **91** more than x . Which equation defines f ?

- A. $f(x) = x^2 + x + 91$
- B. $f(x) = x^2 + 91$
- C. $f(x) = x^2 + 91x$
- D. $f(x) = x^2 + 91x + 91$

ID: 8462b105 Answer

Correct Answer:

C

Rationale

Choice C is correct. It’s given that the function f gives the product of a number, x , and a number that is 91 more than x . A number that is 91 more than x can be represented by the expression $x + 91$. Therefore, f can be defined by the equation $f(x) = x(x + 91)$, or $f(x) = x^2 + 91x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

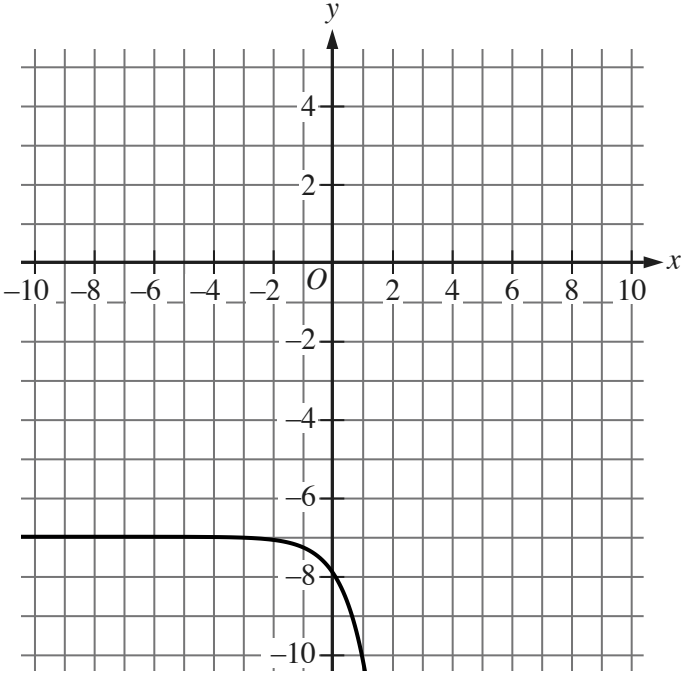
Question Difficulty:

Hard

Question ID df71424b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■ ■ ■

ID: df71424b



The graph of $y = f(x)$ is shown, where $f(x) = ab^x + c$, and a , b , and c are constants. For how many values of x does $f(x) = 0$?

- A. Three
- B. Two
- C. One
- D. Zero

ID: df71424b Answer

Correct Answer:

D

Rationale

Choice D is correct. Each point x, y on the graph of $y = f(x)$ in the xy -plane gives a value of x and its corresponding value of $f(x)$, or y . For any value of x for which $f(x) = 0$, there is a corresponding point on the graph of $y = f(x)$ with a y -coordinate of 0. A point with a y -coordinate of 0 is a point where the graph intersects the x -axis. It's given that $fx = ab^x + c$, where a , b , and c are constants. In the xy -plane, the graph of an equation of this form will lie entirely either above or below the horizontal line defined by $y = c$. The part of the graph of $y = f(x)$ shown lies entirely below the horizontal line defined by $y = -7$, and thus the entire graph of $y = f(x)$ must lie below the line defined by $y = -7$. It follows that the graph of $y = f(x)$ will never intersect the x -axis. Therefore, there are zero values of x for which $f(x) = 0$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty:

Hard

Question ID 668f1863

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 668f1863

Function f is a quadratic function where $f(-20) = 0$ and $f(-4) = 0$. The graph of $y = f(x)$ in the xy -plane has a vertex at $(r, -64)$. What is the value of r ?

ID: 668f1863 Answer

Correct Answer:

-12

Rationale

The correct answer is -12. It's given that function f is a quadratic function where $f(-20) = 0$ and $f(-4) = 0$. It follows that the graph of $y = f(x)$ in the xy -plane passes through the points $(-20, 0)$ and $(-4, 0)$. When the graph of a quadratic function contains two points $(a, 0)$ and $(b, 0)$, the x -coordinate of the vertex of the graph is the average of a and b . Therefore, the x -coordinate of the vertex of the graph of $y = f(x)$ is $\frac{-20 + -4}{2}$, or -12. It's given that the graph of $y = f(x)$ in the xy -plane has a vertex at $(r, -64)$. It follows that the value of r is -12.

Question Difficulty:

Hard

Question ID f25a34aa

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: f25a34aa

The area of a triangle is equal to x^2 square centimeters. The length of the base of the triangle is $2x + 22$ centimeters, and the height of the triangle is $x - 10$ centimeters. What is the value of x ?

ID: f25a34aa Answer

Correct Answer:

110

Rationale

The correct answer is 110. The area of a triangle is equal to one half of the product of the length of the base of the triangle and the height of the triangle. It's given that the length of the base of the triangle is $2x + 22$ centimeters and the height of the triangle is $x - 10$ centimeters; therefore, its area is $\frac{1}{2}(2x + 22)(x - 10)$ square centimeters. It's also given that the area of the triangle is equal to x^2 square centimeters. Therefore, it follows that $\frac{1}{2}(2x + 22)(x - 10) = x^2$. This equation can be rewritten as $(x + 11)(x - 10) = x^2$, or $x^2 + x - 110 = x^2$. Subtracting x^2 from both sides of this equation yields $x - 110 = 0$. Adding 110 to both sides of this equation yields $x = 110$. Therefore, the value of x is 110.

Question Difficulty:

Hard

Question ID a58232b7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: a58232b7

The functions g and h are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?

I. $g(x) = 18(1.16)(1.4)^{x+2}$

II. $h(x) = 18(1.4)^{x+4}$

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

ID: a58232b7 Answer

Correct Answer:
D

Rationale

Choice D is correct. A function defined by an equation in the form $fx = ab^{x+h}$, where a , b , and h are positive constants and $x \geq 0$, has a minimum value of $f(0)$. It's given that function g is defined by $gx = 181.161.4^{x+2}$, which is equivalent to $gx = 20.881.4^{x+2}$. Substituting 0 for x in this equation yields $g0 = 20.88(1.4)^{0+2}$, or $g(0) = 40.9248$. Therefore, the minimum value of $g(x)$ is 40.9248, so $gx = 181.161.4^{x+2}$ doesn't display its minimum value as a constant or coefficient. It's also given that function h is defined by $hx = 181.4^{x+4}$. Substituting 0 for x in this equation yields $h0 = 181.4^{0+4}$, or $h(0) = 69.1488$. Therefore, the minimum value of $h(x)$ is 69.1488, so $hx = 181.4^{x+4}$ doesn't display its minimum value as a constant or coefficient. Therefore, neither I nor II displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Hard